

Weekly Internship Report (20/6/26 - 26/6/26)

WEEK 6

Intern's Name: Atharv Gupta

Internship Week: Week 6 (20/6/26 - 26/6/26)

Company/Organization Name: Red Kross Research Foundation

Mentor's Name: Shivam Mittal, Ojas Satardekar, Piyush Babele

Week Overview:

- Conducted research and demonstrations on OSINT People Investigation tools including OSINT Industries, Intelligence X, Melissa Lookups, PeopleFinders & 411, and Reverse Phone Lookup tools.
- Learned digital forensics concepts including crime scene search methodologies, evidence collection, forensic tools, and live vs. dead forensics.
- Researched advanced cybersecurity concepts covering Active Directory attacks, network enumeration, privilege escalation, credential attacks, Metasploit, reverse shells, password cracking, and defensive security techniques.
- Installed and configured Autopsy Digital Forensics software and explored its features, workflows, and practical forensic analysis capabilities.

Learning Outcomes:

- Learned people's investigation techniques using OSINT tools.
- Improved understanding of digital footprint analysis and identity verification.
- Gained knowledge of cyber forensic investigation processes and evidence handling.
- Developed understanding of Active Directory attack techniques and enterprise security concepts.
- Learned privilege escalation, credential attacks, reverse shells, and security mitigation strategies.
- Gain hands-on experience with Autopsy installation, configuration, and digital forensic analysis.

Challenges Faced:

- Understanding advanced Active Directory attack concepts and their attack flow.
- Learning multiple forensic tools and evidence collection methodologies.
- Research enterprise-level cybersecurity concepts and defensive techniques.

Goals for Next Week:

- Work on the **Threat Intelligence Research Project**.
- Strengthen threat analysis and intelligence documentation skills.
- Improve **Reporting, Documentation, and Executive Communication**.
- Learn how to present technical cybersecurity findings in a clear and professional manner.
- Continue enhancing cybersecurity research and analytical skills.

Mentor Feedback and Suggestions:

- Continue strengthening practical research and analytical skills.
- Focus on understanding enterprise attack techniques and defensive strategies.
- Maintain consistent documentation of cybersecurity learning activities.

Additional Notes:

- This week provided extensive exposure to OSINT investigations, digital forensics, Active Directory security concepts, enterprise attack methodologies, and forensic analysis tools.

Student Diary (Log) Entry

Week	Task Assigned	Activities Performed	Key Learnings	Additional Remarks
1	Cybersecurity Fundamentals and Research Tasks	Malware analysis, GitHub repository setup, Dridex research, Steganography challenge, Cryptography exercises, AI SOC and ZTNA discussions	Learned malware analysis, hashing vs encryption, cryptographic workflows, GitHub documentation, and cybersecurity research concepts	Improved practical understanding of cybersecurity operations and research methodologies
2	Threat Intelligence, Container Security and Web Security Research	Studied MD5 hash extraction, Threat Intelligence concepts, Docker and Docker Networking, Kubernetes basics, Ransomware workflow, OSINT and CSINT, Threat Intelligence Framework, GHDB research, Burp Suite and FoxyProxy, and performed XXE Injection vulnerability analysis using Burp Suite	Learned threat intelligence concepts including CVE, TTPs, Pyramid of Pain, threat actors, OSINT/CSINT methodologies, container technologies, web vulnerability assessment, and XXE exploitation techniques	Enhanced practical cybersecurity research, threat analysis, reconnaissance, and web application security skills
3	Web Application Security Assessment and Penetration Testing	Network Security Assessment, OWASP ZAP security testing, Prototype Pollution labs, Web LLM Attack labs, and web application penetration testing exercises	Learned prototype pollution, DOM XSS, OWASP ZAP usage, Web LLM attack concepts, and penetration testing methodologies	OWASP ZAP usage, Web LLM attack concepts, and penetration testing methodologies Strengthened practical web application security and vulnerability assessment skills
4	OSINT, Threat Intelligence and Security Research	Researched OSINT tools, lock mechanisms, phishing analysis, OpenDNS phishing quiz, Metasploitable environment, vulnerability report research, and GRC concepts	Learned OSINT reconnaissance, data leakage identification, phishing detection, vulnerable environment analysis, and enterprise security governance	Improved analytical and cybersecurity research capabilities
5	Network Security Testing, Web Application Security Testing & CVE Research	Performed Metasploitable FTP/Telnet testing, DVWA vulnerability assessment, SQL Injection testing, and researched 20 recent CVEs	Learned service enumeration, vulnerability assessment, SQL Injection testing, CVE analysis, and security reporting methodologies	Improved practical penetration testing and cybersecurity research skills
6	OSINT Investigation, Digital Forensics & Enterprise	Researched OSINT investigation tools, studied digital forensics concepts, researched Active	Learned OSINT investigation, digital forensics, Active Directory security concepts, privilege	Enhanced enterprise cybersecurity knowledge, digital forensics skills, and

	Cybersecurity Research	Directory attack techniques, installed and configured Autopsy, and prepared cybersecurity documentation	escalation, credential attacks, and forensic analysis using Autopsy	practical security research capabilities
--	------------------------	---	---	--